1. Objectives

The goal is to ensure that the core functionality of the SauceDemo e-commerce website, including login, product browsing, cart management, and checkout, meets the expected functionality and usability standards.

• Verify authentication and login validation across all specified user types, using the expected outcome for each account.

• Ensure accurate addition and removal of cart items, correct cart badge counts, and successful checkout with valid information.

• Verify consistent functional behavior, layout, and navigation across the latest versions of Chrome, Firefox, and Edge.

2. Approach

The test strategy combines manual and automated testing.

• **Manual testing:** UI validation, usability checks, exploratory testing, and functional verification.

• **Automated testing:** Selenium WebDriver regression tests for critical workflows, including login and checkout.

• **Test types:** Functional, UI, input validation, end-to-end, and cross-browser compatibility testing.

• **Input coverage:** Valid and invalid credentials, empty login fields, required checkout fields, and relevant cart states.

Execution and reporting

Execute tests against defined expected results. Record actual results as Pass, Fail, Blocked, or Not Run. Log reproducible defects in Jira and include coverage, execution counts, and unresolved issues in the final summary.

Performance and load testing, security testing, and real payment processing are excluded. Functional testing of performance_glitch_user remains in scope.

3. Scope

In Scope

• **User authentication:** Login with standard_user, locked_out_user, problem_user, performance_glitch_user, error_user, and visual_user.

• **Login validations:** Invalid credentials, empty fields, and the locked-out user error message.

• **Logout:** Logout through the side menu.

• **Products page:** Product details display and sorting by name and price.

• **Product details page:** Open a product and return to the products page.

• **Shopping cart:** Add and remove items, verify the badge count, and validate cart persistence.

• **Checkout workflow:** Your Information, Overview with item total, tax, and total, and Complete.

• **Side menu:** All Items, About navigation, Logout, and Reset App State.

• **UI and usability:** Layout, images, buttons, and navigation.

• **Cross-browser compatibility:** Chrome, Firefox, and Edge.

Out of Scope

• Performance and load testing.

• Security and penetration testing.

• Real payment processing, as the website is a demo.

• Backend, database, and API testing.

• External destination content, including the About page and footer social media links.

• Safari and older browser versions.

• Native mobile applications.

4. Test Environment

Item	Details
Application URL	https://www.saucedemo.com
Environment Type	Remote (public demo website)
Operating System	Windows 10/11
Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)
Device	Desktop/Laptop (1920 x 1080)
Network	Stable Wi-Fi / broadband internet connection
Test Data	Usernames: standard_user, locked_out_user, problem_user, performance_glitch_user, error_user, visual_user. Password for all users: secret_sauce. Checkout: any valid First Name, Last Name, and Postal Code.
Automation Tools	Selenium WebDriver
Test Management & Bug Tracking	Jira

5. Dependencies

- **Website Availability:** SauceDemo must be accessible during testing.
- **Internet Connection:** A stable internet connection is required.
- **Test Accounts:** Valid test credentials are needed to test login behavior.
- **Browsers:** Supported versions of Chrome, Firefox, and Edge must be available.
- **Test Data:** Required product and checkout data must be prepared before execution.

6. Test Deliverables

The following deliverables will be prepared during manual and automated testing of SauceDemo:

Deliverable	Content
Test Scenarios	Main features to be tested, including Login, Product Browsing, Shopping Cart, and Checkout.
Test Cases	Detailed steps, preconditions, test data, and expected results for each scenario.
Test Execution Results	Actual results and execution status for each case: Pass, Fail, Blocked, or Not Run.
Bug Reports	Defect descriptions, severity, steps to reproduce, expected and actual results, and supporting screenshots.
Automation Scripts & Results	Selenium WebDriver regression scripts and execution reports for selected critical workflows.
Test Summary Report	Final coverage, passed and failed case counts, blocked or unexecuted tests, identified bugs, and unresolved issues.

Frequency

• Test scenarios and test cases will be prepared before testing.

• Test results, automation results, and bug reports will be updated during testing.

• The test summary report will be prepared at the end of the test cycle.

Distribution

All test deliverables will be shared with QA team members through a shared folder. Defects will be tracked in Jira.

Formats

Use shared spreadsheets or documents for scenarios, cases, and execution records; Jira for defects; Selenium script files and test runner reports for automation; and a PDF or document for the final summary.

7. Risk Management

The following risks may affect SauceDemo testing. Each risk will be monitored, and mitigation actions will be taken as needed.

Risk	Impact	Mitigation
Website unavailable	Testing may be delayed or blocked.	Check availability before execution, record blocked tests, and resume when the website becomes accessible.
Unstable internet connection	Tests may fail or produce unreliable results.	Use a stable connection and rerun affected tests after connectivity is restored.
Unexpected behavior from special user accounts	Intended account behavior may be reported as a defect.	Define expected results for each user type and compare affected flows with standard_user before reporting defects.
Cart or session data remains between tests	Previous test data may affect later results.	Establish a clean starting state before each independent test. Preserve state intentionally for cart persistence tests.
Limited testing time or tester availability	Some planned tests may remain incomplete.	Prioritize login, cart management, and checkout; document any untested areas in the summary report.
Browser differences or Selenium failures	Tests may fail on specific browsers or produce misleading automation results.	Record browser versions, investigate failures, and manually verify affected flows.
Website changes during testing	Test cases or automation scripts may become outdated.	Review affected cases, update scripts, and rerun relevant regression tests.

Monitoring: Risks and blocked tests will be reviewed daily during test execution. Any impact on scope, coverage, or completion dates will be communicated to the QA team and recorded in the test summary report.

8. Test Schedule

Testing Phase / Activity	Description & Scope	Timeline
Test Planning & Analysis	Review functional requirements, analyze SauceDemo features (Login, Inventory, Cart, Checkout), and define test scope.	Day 1 - Day 2
Test Design & Case Creation	Write manual test cases, prepare test data sets, and outline automation test scripts for critical user paths.	Day 3 - Day 4
Test Environment Setup	Configure target browsers (Chrome, Firefox, Edge) and verify baseline connectivity to the SauceDemo web platform.	Day 5
Test Execution (Phase 1)	Execute functional, UI, and end-to-end test suites. Log discovered defects with detailed reproduction steps.	Day 6 - Day 7
Defect Retesting & Regression	Retest resolved bug fixes and perform regression testing on core workflows to prevent side effects.	Day 8
Test Closure & Reporting	Evaluate exit criteria, aggregate testing metrics, and compile the final Test Summary Report.	Day 9

9. Roles and Responsibilities

Role	Key Responsibilities
Test Manager / Lead	Define overall testing strategy and sign off on the Test Plan document. Monitor progress against milestone deadlines and manage project risks. Coordinate status reporting with core project stakeholders.
QA Engineer / Tester	Design, review, and execute manual test cases across SauceDemo modules. Identify, log, and track software defects in the issue management tool. Perform re-testing of fixed defects and execute regression suites.
Automation Engineer	Develop and maintain automated UI test scripts for critical flows (Login, Checkout). Integrate automated test runs into the continuous testing framework.
Developer	Investigate and resolve reported defects in a timely manner. Deploy updated builds to the dedicated test environment for verification.

Communication Channels & Protocols

Daily Standup: 15-minute daily sync meeting to discuss completed tasks, planned work, and blockers.

Bug Tracking Protocol: All defects must be logged in Jira/GitHub Issues with clear severity levels, screenshots, and steps to reproduce. Jira is the primary tool specified in this plan.

Primary Communication: Dedicated Slack / Teams channel for real-time team collaboration and urgent escalation items.

Public demo constraint: Developer fixes, updated builds, and retesting depend on access to the site maintainers; the QA team cannot deploy changes to the public demo.